

Day 11 – File Ownership Challenge (chown & chgrp)

Task 1: Understanding Ownership

Check ownership in home directory

```
ls -l ~
```

Format explained:

-rw-r--r-- 1 owner group size date filename

- **Owner** → user who owns the file
- **Group** → group that shares access

 Screenshot: ls -l output

Document (answer):

- Owner controls the file by default
 - Group allows controlled sharing with other users
-

Task 2: Basic chown Operations

Create file

```
touch devops-file.txt  
ls -l devops-file.txt
```

Change owner to tokyo

```
sudo chown tokyo devops-file.txt  
ls -l devops-file.txt
```

Change owner to berlin

```
sudo chown berlin devops-file.txt  
ls -l devops-file.txt
```

 Screenshot: before & after ownership

Task 3: Basic chgrp Operations

Create file

```
touch team-notes.txt  
ls -l team-notes.txt
```

Create group

```
sudo groupadd heist-team
```

Change group

```
sudo chgrp heist-team team-notes.txt  
ls -l team-notes.txt
```

 Screenshot: group change

Task 4: Combined Owner & Group Change

Create file

```
touch project-config.yaml
```

Change owner + group (one command)

```
sudo chown professor:heist-team project-config.yaml  
ls -l project-config.yaml
```

Directory ownership

```
mkdir app-logs  
sudo chown berlin:heist-team app-logs  
ls -ld app-logs
```

 Screenshot: file + directory ownership

Task 5: Recursive Ownership

Create structure

```
mkdir -p heist-project/vault  
mkdir -p heist-project/plans  
touch heist-project/vault/gold.txt  
touch heist-project/plans/strategy.conf
```

Create group

```
sudo groupadd planners
```

Recursive ownership change

```
sudo chown -R professor:planners heist-project/
```

Verify

```
ls -lR heist-project/
```

 Screenshot: recursive ownership

Task 6: Practice Challenge (Real-World Simulation)

Create users (if not already present)

```
sudo useradd -m tokyo
sudo useradd -m berlin
sudo useradd -m nairobi
```

Create groups

```
sudo groupadd vault-team
sudo groupadd tech-team
```

Create directory & files

```
mkdir bank-heist
touch bank-heist/access-codes.txt
touch bank-heist/blueprints.pdf
touch bank-heist/escape-plan.txt
```

Set ownership

```
sudo chown tokyo:vault-team bank-heist/access-codes.txt
sudo chown berlin:tech-team bank-heist/blueprints.pdf
sudo chown nairobi:vault-team bank-heist/escape-plan.txt
```

Verify

```
ls -l bank-heist/
```

 Screenshot: final ownership state

Documentation File

Create **day-11-file-ownership.md**

Day 11 Challenge – File Ownership

Files & Directories Created

- devops-file.txt
- team-notes.txt
- project-config.yaml
- app-logs/
- heist-project/
- bank-heist/

Ownership Changes

- devops-file.txt: user → tokyo → berlin
- team-notes.txt: group → heist-team
- project-config.yaml: professor:heist-team
- heist-project/: professor:planners (recursive)
- bank-heist files:
 - access-codes.txt → tokyo:vault-team
 - blueprints.pdf → berlin:tech-team
 - escape-plan.txt → nairobi:vault-team

Commands Used

- ls -l, ls -lR
- chown, chgrp
- chown -R
- useradd, groupadd

What I Learned

- Difference between file owner and group
- How to safely change ownership
- Why recursive ownership is critical for applications

Troubleshooting Quick Fixes

Permission denied

sudo chown user:group file

Group doesn't exist

sudo groupadd groupname

User doesn't exist

```
sudo useradd username
```